

```

# Program to demonstrate Laws of Sets
# User selects number of sets and the law to verify

# =====
# TAKE NUMBER OF SETS
# =====

while True:
    n = int(input("How many sets do you want to enter (1-3)? "))

    if 1 <= n <= 3:
        break
    else:
        print("Please enter a number between 1 and 3.")

# =====
# TAKE SETS FROM USER
# =====

sets = {}

for i in range(n):
    elements = input(
        f"Enter elements of Set {chr(65 + i)} separated by spaces: "
    )

    sets[chr(65 + i)] = set(map(int, elements.split()))

# Assign sets
A = sets["A"]

if n >= 2:
    B = sets["B"]

if n >= 3:
    C = sets["C"]

# =====
# UNIVERSAL SET
# =====

U_input = input(
    "\nEnter elements of Universal Set U separated by spaces: "
)

U = set(map(int, U_input.split()))

E = set()

# =====
# COMMUTATIVE LAW
# =====

def commutative_law():

    if n < 2:
        print("\nCommutative Law requires at least 2 sets.")
        return

    print("\n--- Commutative Law ---")

    # Union
    left = A | B
    right = B | A

    print("A ∪ B =", left)
    print("B ∪ A =", right)
    print("A ∪ B = B ∪ A :", left == right)

    # Intersection
    left = A & B
    right = B & A

    print("\nA ∩ B =", left)
    print("B ∩ A =", right)
    print("A ∩ B = B ∩ A :", left == right)

```

```

print("\n--- Union Law ---")

# =====
# ASSOCIATIVE LAW
# =====

def associative_law():

    if n < 3:
        print("\nAssociative Law requires 3 sets.")
        return

    print("\n--- Associative Law ---")

    # Union
    left = (A | B) | C
    right = A | (B | C)

    print("(A U B) U C =", left)
    print("A U (B U C) =", right)
    print("Union Associative Law :", left == right)

    # Intersection
    left = (A & B) & C
    right = A & (B & C)

    print("\n(A n B) n C =", left)
    print("A n (B n C) =", right)
    print("Intersection Associative Law :", left == right)

# =====
# DISTRIBUTIVE LAW
# =====

def distributive_law():

    if n < 3:
        print("\nDistributive Law requires 3 sets.")
        return

    print("\n--- Distributive Law ---")

    # A U (B n C)
    left = A | (B & C)
    right = (A | B) & (A | C)

    print("A U (B n C) =", left)
    print("(A U B) n (A U C) =", right)
    print("Union Distributive Law :", left == right)

    # A n (B U C)
    left = A & (B | C)
    right = (A & B) | (A & C)

    print("\nA n (B U C) =", left)
    print("(A n B) U (A n C) =", right)
    print("Intersection Distributive Law :", left == right)

# =====
# IDENTITY LAW
# =====

def identity_law():

    print("\n--- Identity Law ---")

    # A U ∅ = A
    left = A | E
    right = A

    print("A U ∅ =", left)
    print("A =", right)
    print("A U ∅ = A :", left == right)

    # A n U = A
    left = A & U
    right = A

    print("\nA n U =", left)
    print("A =", right)

```

```

print("A  $\cap$  U = A :", left == right)

# =====
# COMPLEMENT LAW
# =====

def complement_law():

    print("\n--- Complement Law ---")

    A_comp = U - A

    print("A =", A)
    print("A' =", A_comp)

    # A  $\cup$  A' = U
    left = A | A_comp
    right = U

    print("\nA  $\cup$  A' =", left)
    print("U =", right)
    print("A  $\cup$  A' = U :", left == right)

    # A  $\cap$  A' =  $\emptyset$ 
    left = A & A_comp
    right = E

    print("\nA  $\cap$  A' =", left)
    print(" $\emptyset$  =", right)
    print("A  $\cap$  A' =  $\emptyset$  :", left == right)

# =====
# DE MORGAN'S LAWS
# =====

def demorgan_law():

    if n < 2:
        print("\nDe Morgan's Laws require at least 2 sets.")
        return

    print("\n--- De Morgan's Laws ---")

    A_comp = U - A
    B_comp = U - B

    # First De Morgan's Law
    # (A  $\cup$  B)' = A'  $\cap$  B'

    left = U - (A | B)
    right = A_comp & B_comp

    print("(A  $\cup$  B)' =", left)
    print("A'  $\cap$  B' =", right)
    print("(A  $\cup$  B)' = A'  $\cap$  B' :", left == right)

    # Second De Morgan's Law
    # (A  $\cap$  B)' = A'  $\cup$  B'

    left = U - (A & B)
    right = A_comp | B_comp

    print("\n(A  $\cap$  B)' =", left)
    print("A'  $\cup$  B' =", right)
    print("(A  $\cap$  B)' = A'  $\cup$  B' :", left == right)

# =====
# MAIN PROGRAM
# =====

print("\n=====")
print("          LAWS OF SETS")
print("=====")

print("\nEntered Sets:")

for name, value in sets.items():
    print(f"Set {name} =", value)

print("Universal Set U =", U)

```

```
while True:

    print("\n=====")
    print("Select a Law")
    print("=====")

    print("1. Commutative Law")
    print("2. Associative Law")
    print("3. Distributive Law")
    print("4. Identity Law")
    print("5. Complement Law")
    print("6. De Morgan's Laws")
    print("7. All Laws")
    print("8. Exit")

    choice = int(input("\nEnter your choice: "))

    if choice == 1:
        commutative_law()

    elif choice == 2:
        associative_law()

    elif choice == 3:
        distributive_law()

    elif choice == 4:
        identity_law()

    elif choice == 5:
        complement_law()

    elif choice == 6:
        demorgan_law()

    elif choice == 7:
        commutative_law()
        associative_law()
        distributive_law()
        identity_law()
        complement_law()
        demorgan_law()

    elif choice == 8:
        print("\nProgram ended.")
        break

    else:
        print("\nInvalid choice! Please enter 1 to 8.")
```

```
--- De Morgan's Laws ---
(A ∪ B)' = {8, 9, 6, 7}
A' ∩ B' = {8, 9, 6, 7}
(A ∪ B)' = A' ∩ B' : True

(A ∩ B)' = {1, 5, 6, 7, 8, 9}
A' ∪ B' = {1, 5, 6, 7, 8, 9}
(A ∩ B)' = A' ∪ B' : True

=====
Select a Law
=====
1. Commutative Law
2. Associative Law
3. Distributive Law
4. Identity Law
5. Complement Law
6. De Morgan's Laws
7. All Laws
8. Exit

Enter your choice: 8

Program ended.
```